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ARTICLE

Enhancing Student Feedback in Data Science Education: Harnessing the Power of AI-Generated Approaches

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Abstract
Data science has emerged as an important field in higher education, presenting complex challenges for students due to its interdisciplinary nature involving statistical and computational concepts. In previous research, we introduced the rDSA system (R Data Science Assignments) to support data science students during formative assessments by providing exercises and feedback. As known, formative assessment has evolved from an "interim" evaluation method into a continuous process that enhances student understanding and informs educators to adjust their teaching strategies. It is closely linked to scaffolding, i.e., the method aiming to guide students from their current knowledge to the skills they can develop next. The research reported in this paper builds on previous findings that showed improved learning outcomes and good student engagement. We explored how to enhance student feedback by incorporating Large Language Models (LLMs) with traditional AI into the rDSA system. The results show that students extensively used the system, submitting numerous solutions and attempting various exercises, with activity peaking before exams. The system exceeded expectations in providing useful insights and feedback. However, students faced challenges related to focused attention, probably due to the verbosity of feedback. Learning outcomes, on average, improved, with students who solved more exercises using the tool, performing better than the others.

Keywords Generative AI · Large language models · LLMs · Automated feedback · Data science

Introduction

Data science has become a significant field within higher education in recent years. Mastering the subject poses complex challenges for students because it demands the

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assimilation, integration, and application of statistical and computational principles (e.g., data collection, data analysis, inference, algorithms, machine learning) (Buckingham Shum et al., 2013). In addition, students who attend data science courses have varying preparation, culture, and scholastic backgrounds (Schwab-McCoy et al., 2021; Brooks et al., 2021). Deploying effective teaching strategies and tools to aid data science students is, therefore, a relevant and ongoing challenge.

In previous research, we designed and developed the rDSA system (R Data Science Assignments) that supported our students during formative assessment activities by providing exercises and automated feedback on their solutions (Bernardi et al., 2019; Vittorini et al., 2020; Vittorini & Galassi, 2022). As known, formative assessment was initially introduced as an "interim" evaluation tool for educators and learners, that evolved into an ongoing process offering insights into student understanding, helping to adapt teaching approaches (Scriven, 1967; Bennett, 2011; Black & Wiliam, 2010). Formative assessment is closely linked to scaffolding because both aim to guide learners from their current knowledge to the skills they can develop next (Shepard, 2005; Kruiper et al., 2022), i.e., in the zone of proximal development (Vygotsky, 1978). Yet, while formative assessment adjusts teaching based on insights into the learner's comprehension, scaffolding offers support through hints and motivation. According to studies, feedback for formative assessment and scaffolding supports the building of new knowledge throughout the whole learning process and has the potential to improve the learning outcomes and self-regulated learning strategies (Frank et al., 2018; Zuo et al., 2023; Kingston & Nash, 2011; Zimmerman, 1990).

In Vittorini et al. (2020), we showed that our feedback improved the learning outcomes and was used by students who expressed good engagement while using it. The recent progress in Large Language Models (LLMs) and the encouraging results that started to appear about the potential impact of feedback generated by LLMs (e.g., Dai et al., 2024; Zhang et al., 2024; Lyu et al., 2024) pushed us to evaluate the possibility to improve our system through the use of LLMs. Built using deep learning techniques, particularly neural networks, LLMs are trained on vast text corpora to capture various language components (including syntax, semantics, and cultural context) and generate language that resembles human communication. LLMs represent part of Generative AI, i.e., the set of systems capable of creating new content—such as text, images, or audio—by leveraging patterns and examples learned during their training (Raiaan et al., 2024).

In this paper, we initially discuss the improvements introduced in the student feedback, i.e., through a toolchain that exploits both traditional AI and LLMs. Then, we report on the system use and the impact it had on student engagement and academic performance in several courses held at the University of L'Aquila, Italy. Specifically, we address the following research questions:

The paper is organized as follows. We start with the related work on feedback (Section "Related Work"). Then, we present the theoretical framework for our research by describing formative assessment, scaffolding, feedback, and their effectiveness (Section "Theoretical Framework"). Hence, we show an applicative scenario that describes the types of exercises provided by the system and the feedback students

receive. Starting from these premises, the paper focuses on the experimental results: Section "Materials and Methods" describes the material and methods, Section "Results" summarizes the results by each research question, and Section "Discussion" discusses and relates them with the existing literature. Finally, the paper ends with the conclusions, where the study's limitations and future work are also presented (Section "Conclusions, Limitations and Future Work").

Related Work

Feedback Evolution in Education

The evolution of feedback in educational contexts has undergone a significant change over the past few decades. Early forms of simple feedback, such as Knowledge of Results (KR), developed in the 1960s, were usually confined to binary responses, merely indicating whether a task was correct or incorrect. A pioneering example is the automated correction system introduced by Hollingsworth (Hollingsworth, 1960), which focused mainly on verifying the syntactic correctness of code. These systems proved useful for learning basic skills, but they lacked the ability to provide more elaborate guidance or to promote critical reflection and deep understanding on the part of the student.

Subsequent technological advancements have led to the development of more sophisticated feedback mechanisms, particularly in programming courses. For instance, feedback systems employing static and dynamic code analysis techniques have become increasingly prevalent (Keuning et al., 2018). Static analysis allows for detecting incorrect syntactic structures or concepts without executing the code, while dynamic analysis compares the output of executed code with the expected result. These approaches have paved the way for more advanced types of feedback, such as Knowledge of Correct Response (KCR), which provides more informative insights than KR, though still limited to confirming correctness or revealing the mistake. From these foundations, Elaborated Feedback (EF) has emerged as a more sophisticated solution, capable of providing in-depth explanations regarding students' correct or incorrect answers. For instance, Diri Koç and Öner (2023) showed that EF can be particularly useful in complex contexts, as it allows students to connect new knowledge to previous knowledge, promoting the development of advanced cognitive skills.

However, EF can lead to cognitive overload if not properly managed, especially for students with limited basic skills (Lipnevich & Panadero, 2021). Additional details or elaborate feedback, such as narrative explanations, may overload students' working memory, making it difficult for them to integrate and apply the advice received. This phenomenon is related to the principles of Cognitive Load Theory (CLT) and becomes particularly problematic for students with fewer basic skills or those unable to handle multiple feedback cycles simultaneously (Sweller et al., 2011). Similarly, the study of Wang et al. (2019) found that detailed explanations stimulated learners' perception of feedback, decreased the cognitive load due to ineffective instructional guidance, and increased learning motivation. On the other hand, a study by Lang et al. (2022)

showed that the mediation of a pedagogical agent (PA)[1] to provide EF enhanced intrinsic motivation and transfer performance while reducing the cognitive processes related to learning.

Feedback Customization with AI

The introduction of technologies based on artificial intelligence (AI) and machine learning (ML) has improved adaptive feedback systems, making them capable of providing personalized real-time responses.

Intelligent Tutoring Systems (ITSs) use AI to monitor student progress and dynamically adjust feedback. Research has shown that these systems can enhance learning, for instance, for the personalization of support according to students' cognitive and metacognitive needs (Roll et al., 2011), or to produce the list of concepts that a student is ready to learn (Falmagne et al., 2013). In addition, Natural Language Processing (NLP) technologies have enabled a more detailed and nuanced performance assessment and feedback, facilitating a deeper understanding of complex concepts (Burrows et al., 2015).

A recent literature review (Deeva et al., 2021) underlines the importance of automated feedback for learners, as well as its effectiveness due to the recent advances of computer technologies achieved in the last decades. In particular, Generative AI and LLMs have given an additional push, showing abilities and performances that seem to open new educational opportunities (Yan et al., 2024). Accordingly, researchers started to explore the possibility of using Generative AI in education from different perspectives.

Hereafter, we summarize recent research about adopting LLMs to provide feedback in computer science, in other domains, and related pedagogical issues.

LLM-feedback in Computer Science

Several attempts exist to exploit LLMs in building feedback for courses in computer science.

A study close to our research (and particularly for RQ_1) was conducted by Gabbay and Cohen (2024). The authors examined the quality of feedback generated by LLMs in programming courses delivered through Massive Open Online Courses (MOOC). They compared the LLM feedback with Automated Test-based Feedback (ATF). The study was based on 500 code solutions, and the authors gathered qualitative perceptions from students who used their application. Students positively evaluated the readability of the feedback (3.83/5) but gave lower scores for its usefulness in understanding errors (2.98/5). The main issues highlighted included interpretation errors (hallucinations) and inconsistencies in repeated feedback. However, the authors reported some limitations, like mislabelling incorrect code as correct, sometimes omitting critical error explanations, or including incorrect suggestions in the feedback. Similar results

[1] Pedagogical agents are animated human-like characters used within digital learning platforms to deliver information and boost motivation by mimicking social interactions with students.

are reported in Zhang et al. (2024), where students perceived the feedback as specific and clear (4.02/5).

Jürgensmeier and Skiera (2024) explored using different LLMs to provide automated feedback on complex multimodal exercises requiring coding, statistics, and economic reasoning. The primary objective was to assess the effectiveness and accuracy of AI-generated feedback compared to human expert feedback across 4349 solutions from 243 learners. The study showed that LLMs can deliver correlated and unbiased quantitative feedback, with the best LLM showing a strong correlation (r = 0.94) with human evaluations and minimal deviation (6% mean absolute error). Additionally, the authors surveyed 43 undergraduate learners indicating a high acceptance and recommendation rate for the AI feedback app used, with 93% of respondents willing to use it again and 84% recommending it to peers.

A smaller but still relevant study was conducted with a single graduate-level computer science course at a public University in the United States (Jia et al., 2024). Twenty-eight students answered a survey and reported a prevailing satisfaction with the LLM-generated feedback, which was perceived as helpful and comprehensive. On the other hand, students had concerns about its faithfulness and that it could replace the instructor's feedback.

Another study related to our research, and in particular for RQ_3, is the research from Lyu et al. (2024). The authors introduced an LLM-based system called CodeTutor in a programming course, observing significant improvements in students' final scores compared to a control group. In addition, students positively evaluated CodeTutor's ability to interpret questions and provide syntactic assistance, although they expressed doubts about its effectiveness in fostering critical thinking skills.

In Pahi et al. (2024), the authors describe an active and personalized learning environment supported by in-class exercises. Teaching Assistants (TAs) were flanked by an LLM to monitor and provide immediate feedback to students on the solutions. This method was evaluated in several undergraduate and graduate computer science and data science courses. Among the results, when TAs and the LLM revised each other's feedback to students, the quality of the feedback improved in several aspects, including technical quality and the identification of conceptual gaps, as well as suggesting the next actions for students to take.

LLM-feedback in other Domains

Another education tool that uses LLMs is explored by Haindl and Weinberger (2024). The authors focused on students' perceptions and the effectiveness of AI feedback compared to human feedback. In a six-week course with 91 ENL (English as a New Language) students, the study found that AI-generated feedback is comparable to human feedback, and no significant differences in academic performance emerged between the two groups. Students' preferences show an increasing appreciation for AI feedback, although concerns about comprehensibility and accuracy remain.

Similar results are reported in Escalante et al. (2023). The study first examined the learning outcomes of 48 university English ENL learners with an experimental and control group, which received feedback from an LLM and a human tutor, respectively. The results indicated no significant differences in language progress between

the two groups, suggesting that both feedback methods were equivalent in terms of their effectiveness in improving students' writing skills. In addition, half of the students preferred human feedback (appreciating face-to-face interaction and emotional benefits such as increased engagement and improved oral skills), and the other half preferred AI-generated feedback (citing the clarity and specificity of corrections as particularly helpful in improving writing).

On the other hand, Meyer et al. (2024) reported a positive outcome on how using LLMs to provide feedback in writing classes led to significant improvements in students' revision of their work (i.e., students who received feedback generated by LLM improved on an essay writing task, compared to students who received no feedback), with increased motivation and emotional engagement. These findings directly relate to our research questions RQ_2 and RQ_3.

Moreover, a survey with 462 German-speaking students (studying to be teachers) (Ruwe & Mayweg-Paus, 2024) showed that students tended to perceive LLMs as more reliable than educators, with this perceived reliability linked to factors such as competence, integrity, and cordiality.

Theoretical Framework

As pointed out by Stamper et al. (2024), current feedback systems based on LLMs have pedagogical limitations, e.g., they focus only on immediate performance improvements at the expense of deep and lasting learning, they risk providing inconsistent suggestions or "hallucinations". As a consequence, the authors underline the importance of grounding LLM-based feedback generation systems in theoretical frameworks and evidence-based approaches to improve their learning effectiveness. For instance, in Lin and Chang (2023), the authors integrate personalized chatbots with active and self-regulated learning to improve student involvement, motivation, and educational outcomes. An integration of self-regulated learning and judgement of learning is instead discussed in Chang et al. (2023), so as to devise pedagogical principles useful for integrating AI chatbots in classrooms.

In our research, we designed, developed, and deployed the rDSA system in terms of a theoretical framework that combines formative assessment, scaffolding, and elaborated feedback, applying cautious and theoretically grounded methods coherently with the pedagogical issues mentioned above.

Formative assessment was initially introduced as a technique to provide feedback essential for analyzing learning developments (Scriven, 1967). Over time, it has become an ongoing process offering feedback to educators and students, and helping adapt teaching approaches and learning strategies (Bennett, 2011; Black & Wiliam, 2010). Formative assessment is closely linked to scaffolding (Wood et al., 1976; Clark, 2012; Shepard, 2005; Kruiper et al., 2022), both aiming to guide learners into the zone of proximal development (Vygotsky, 1978), i.e., from what they currently know to what they can acquire next through guidance. Yet, while formative assessment adjusts teaching based on insights into the learner's comprehension, scaffolding offers support through hints and motivation. Adaptive scaffolding, which customizes support to meet individual learner needs, represents the most effective form of scaffolding in

fostering targeted skill development (Azevedo et al., 2004). The meta-analysis discussed in Kingston and Nash (2011) highlights that using computer-based formative assessment systems results in an average effect size of 0.28 on learning outcomes; the meta-analysis from Doo et al. (2020) reports different effect sizes (depending on the specific type of scaffolding technique) ranging from 0.13 to 0.38.

Feedback is a crucial point for both formative assessment and scaffolding (Frank et al., 2018) because it provides the needed guidance on building new knowledge based on the existing understanding of the subject throughout the whole learning process. Various learning theories suggest that feedback is most beneficial to students when: (i) it steers learners towards the correct response (behaviorism); (ii) it is focused on the task, detailed, precise, corrective, and aids in closing the gap between current and desired performance (cognitivism); (iii) it helps identify the discrepancy between present and aimed performance levels (meta-cognitivism); (iv) it is oriented towards tasks or objectives, emphasizes the learning process, is timely, positive, non-critical, constructive, and offers thorough explanations or justifications (social constructivism) Thurlings et al. (2013). Among the different types of feedback, i.e., KR, KCR, and EF, a recent meta-analysis (Mertens et al., 2022) indicates that:

- EF had a greater effect size (0.49) compared to KR (0.05) or KCR (0.32);
- EF proved more effective than KR and KCR in achieving better learning outcomes;
- the effect size was also adversely influenced by delayed feedback timing.

Therefore, the developed system:

- aims to scaffold students during formative assessment;
- includes a timely and constructive EF that explains why the learner's response is correct (or not), allows the learner to review the response until providing the right solution to the exercise, and incorporates thorough explanations using LLMs.

Applicative Scenario

The courses where the rDSA system is used for formative assessment are organized as follows. At the beginning of the course, students are informed about all the tools they can use, namely:

- a dedicated website with all information about the course;
- the slides the professor uses during the course;
- video recordings of past lectures about the course topics (Vittorini & Galassi, 2021);
- the rDSA system, presented as the system that enables students to practice with the course topics through exercises, whose solutions are subject to automated feedback.

An example of an exercise that students could be required to solve during formative assessment is the following.

Consider dataset `sbp`*. The dataset contains data on ten patients. For each patient, it includes the gender (variable* `gender`*), and the systolic blood*

pressure before (variable `before`*) and after (variable* `after`*) taking an anti-hypertensive drug.*
Verify whether the difference in pressure after taking the drug between males and females is statistically significant, and comment on the test result.
Submit as a solution a text that contains the list of R commands with the respective output and the comments given on the results.

To solve the exercise correctly, a student must first identify the study design (it is a two-independent-samples design), then check the normality of the data (the samples are made up of only five patients each, so a normality test is mandatory), accordingly choose between a parametric or non-parametric test, and finally interpret the test result. Let us suppose that a student elaborates the following solution in R (where the student did not perform the required normality tests):

```
> t.test(sbp$after[sbp$gender=="m"], sbp$after[sbp$gender
    ↪ =="f"])

    Welch Two Sample t-test

data:  sbp$after[sbp$gender == "m"] and sbp$after[
    ↪ sbp$gender == "f"]
t = 2.4771, df = 7.0303, p-value = 0.04224
alternative hypothesis: true difference in means is not
    ↪ equal to 0
95 percent confidence interval:
  1.544818 65.255182
sample estimates:
mean of x mean of y
    190.8     157.4

> # Given that the p-value is less than 0.05, I consider
    ↪ the difference in blood pressure statistically
    ↪ significant and, therefore, generalizable to the
    ↪ population.
```

The student copies/pastes the solution from R into the system and then asks for feedback. Two different feedbacks are available, and are depicted in Fig. 1. On the top, there is the *schematic feedback*, based on "traditional" AI, i.e., static code analysis for the commands; embeddings, custom features, and a binary classifier for analyzing the comments (Vittorini et al., 2020; Angelone et al., 2021; Vittorini & Galassi, 2022). On the bottom is the *combined feedback*, based on LLM: it is generated through prompt engineering, exploits the *schematic feedback*, and is based on ChatGPT-4 (Letteri & Vittorini, 2024). Students from the previous academic year used the *schematic feedback*, and students from this academic year used the *combined feedback*.

Figure 2 shows the generation process for both the *schematic* and *combined feedback*. To produce the *schematic feedback*, the system compares the student's solution with the professor's correct solution. It analyzes the commands and the comments one by one, then: (i) checks if they are right or wrong, (ii) provides explanations on the possible reason for the mistake, (iii) proposes changes toward the correct solution, (iv) lists the missing commands, and (v) estimates the final grade, using different colors to separate what is good from what is wrong. To produce the *combined feedback*, the

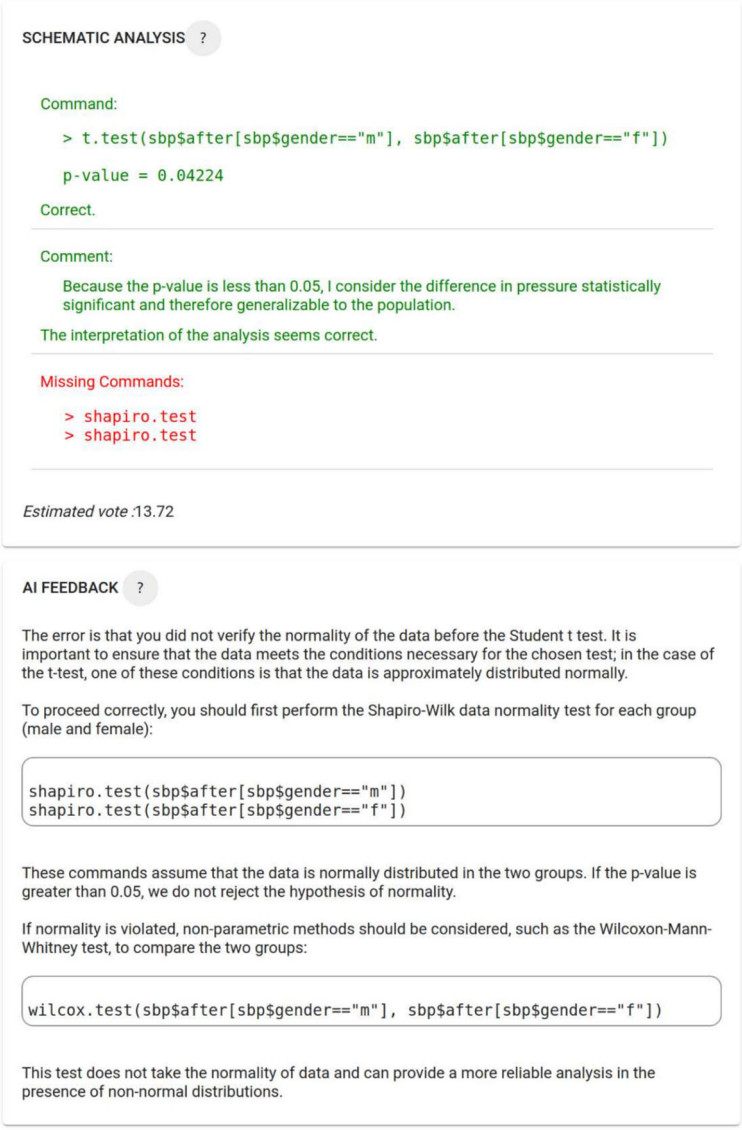


Fig. 1 The two feedback solutions: on top the *schematic feedback*, on the bottom the *combined feedback*

system creates a prompt that asks the LLM to focus on the mistake and how to solve it. The adopted prompt (see Letteri and Vittorini, 2024 for a detailed description and evaluation of the prompt engineering process):

- is built by taking into account the exercise, the solution given by the student, the student model[2] (Vittorini, 2023), and the schematic feedback;

[2] Among all information contained in the student model, the prompt only uses the student's ability in the subject and its topics.

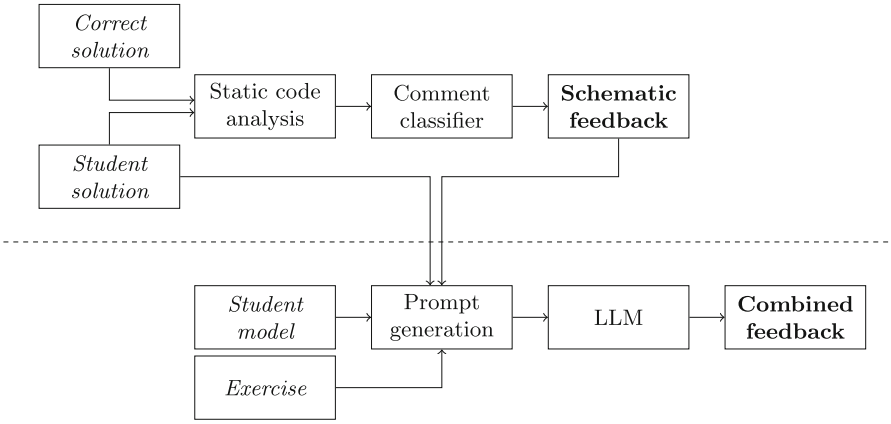


Fig. 2 The generation process for both the *schematic* and *combined feedback*. On the left, in italics, the needed information. In the center, the involved steps. On the right, in bold, the *schematic feedback* and *combined feedback*

- is instructed to produce feedback that initially explains the reason for the mistake and adds how to proceed correctly, first suggesting the correct commands and finally showing the commands that should be issued (because either missed or wrong).

In the specific example of Fig. 1, it is worth remarking that the LLM mentioned the potential command that one should use in case the Shapiro-Wilk test would suggest a non-normal distribution (i.e., the Wilcoxon-Mann-Whitney test), which is not included in the *schematic feedback*.

Regardless of the type of feedback, a student can revise the solution and submit the new one to be analyzed by the system for further feedback. This process can be iterated until the student reaches the correct solution or decides to quit the exercise.

Materials and Methods

The study is organized as shown in Fig. 3. We enrolled two cohorts of students from the two academic years (AY) 2023/24 and 2022/23. The first cohort used the *combined feedback* (based on LLMs), the second cohort used the *schematic feedback* (based on traditional AI). Both sample sizes (136 and 139, respectively) and the gender distribution (71% and 72% of the females, respectively) were analogous (see Table 1). In addition, even if we did not collect quantitative data, the observations made in the classroom did not show evident differences in terms of age distribution. At the same time, potential differences in statistics and computer science background between the two AYs may exist. However, a mitigation aspect is that all the involved degrees have a programmed access, which means that all students at least have the same minimum level of competencies.

To answer RQ_1, we adopted the expectation/experience design (Tullis & Albert, 2013). This design is commonly used in user experience research: it involves first

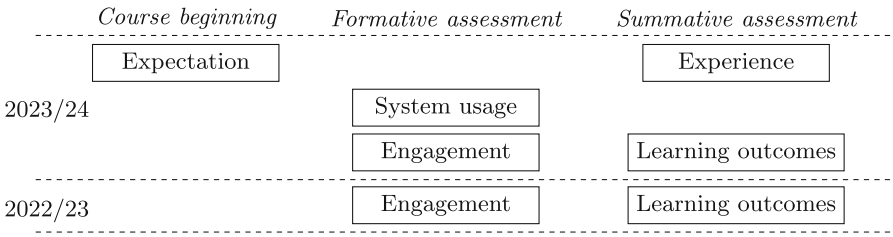


Fig. 3 Study organization. Two cohorts of students (AY 2023/24 and 2022/23), three phases (course beginning, formative assessment, summative assessment), five data sources (expectation, experience, system usage, engagement, and learning outcomes)

asking users about their expectations of a product, service, or feature before interacting with it, and then gathering their experiences after use. By comparing expectations with actual experiences, designers can identify gaps, validate assumptions, and uncover areas for improvement. Accordingly, at the beginning of the course, we asked the students in the first cohort (AY 2023/24) to complete a questionnaire regarding their expectations. The questionnaire was made up of two questions: (i) how easy they expected to understand their progress with the subject, and (ii) how easy they expected to receive constructive feedback able to suggest how to correct their mistakes. At the end of the course, we asked the same students to report on their experience with the system, in terms of the same dimensions investigated for their expectations. This time, obviously, the questions were declined in terms of their experience. For both questionnaires, the answers were coded as a 5-point Likert scale, where 1=Very difficult, 2=Difficult, 3=Average, 4=Easy, and 5=Very easy. The higher the rating, the more satisfactory the expectation or experience. See Appendix A.1 for details.

Concerning RQ_2, we examined the engagement of the students of this AY and compared it with the engagement of the students of the previous one. To measure the engagement, we used the UAES (User Assessment Engagement Scale) questionnaire (Angelone & Vittorini, 2019). It evaluates engagement globally and with four subscales, namely: focused attention (FA), perceived usability (PU), aesthetics (AE), and reward (RW). In particular, FA measures if a student feels absorbed in the assessment and loses track of time; PU assesses the quality of the interaction; AE quantifies the attractiveness and visual appeal of the interface; and RW measures how students feel rewarded by using the system. For all dimensions, the higher the value, the better the perceived engagement. See Appendix A.2. The questionnaire was proposed to the students in both cohorts during the respective courses.

Table 1 Participants. Students in the 2023/24 cohort used *combined feedback* (based on LLM), students in the 2022/23 cohort used *schematic feedback* (based on traditional AI)

AY	N	M	F
2022/2023	139	39 (28%)	100 (72%)
2023/2024	136	39 (29%)	97 (71%)

Regarding RQ_3, we analyzed the students' grades of this AY, explored if they are related to the system usage or the engagement, and compared them with those of the previous one. Grades are in the Italian system, i.e., in a range from 0 to 30 cum laude, customarily considered 31. A passing grade is 18/30.

Finally, we collected the system usage during the formative assessment activities. The system usage included the submitted solutions, the attempted exercises, and the student's ability estimated through the Rasch model (Rasch, 1960). As is known, the Rasch model estimates a person's ability based on their right or wrong answers to questions. It usually ranges from -3 to 3, and it is automatically estimated for our students by the learning analytics engine available in the rDSA system. We use the estimated ability as a precursor of the final grade: during formative assessment, we estimate the ability using the Rasch model to understand the student's progress; at the end of the course, we assign a manual grade to define the student's final proficiency.

Note that to improve comparability between the different AYs, we collected the engagement and grades achieved only by the students in the second cohort who attended the first exam session.

The descriptive analysis used means or medians, with 95% confidence intervals, to calculate the central tendency. The dispersion was instead calculated through standard deviation or interquartile range. We used histograms to plot categorical data. Distributions were analyzed through boxplots: they show the central tendency (the median, as a vertical line), spread (the inter-quartile range, as a box), and the outliers in the data (using dots). Finally, the comparisons were performed with paired (in case of a pre/post design) or independent (in case of independent groups) t-tests.

The students received informed consent and could consult the privacy policy before voluntarily participating in the activities reported in the study.

Results

System Usage

The formative assessment period (i.e., from the beginning of the course until the first exam session) lasted 36 days. During this period, 132 students (97%) used the system. However, usage was not uniform by time or by students.

Figure 4 shows the distribution of students by the number of submitted solutions and solved exercises [3]. We used these two metrics because they measure two different aspects of how students interacted with the system during formative assessment. The number of solved exercises measures the extent of topics a student has covered. The number of submitted solutions assesses the quantity of work a student has performed. Therefore, the two metrics aim to capture the "qualitative" (the number of exercises) and "quantitative" (the number of submitted solutions) interaction with the exercises offered by the system. The median number of submitted solutions is 128, the first quartile is 48, and the third quartile is 228. The analysis shows the presence of eight outliers. In particular, two students submitted more than one thousand solutions to

[3] We considered an exercise solved if at least a solution was given to it.

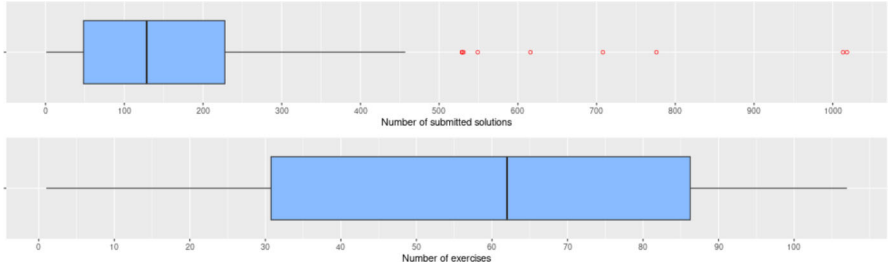


Fig. 4 Distribution of students by number of submitted solutions (top) and exercises (bottom)

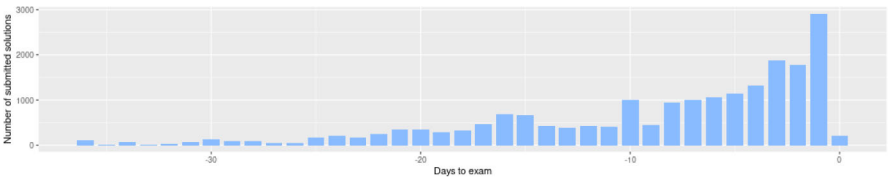
exercises. The distribution of solved exercises is more regular. The median is 62, the first quartile is 31, and the third quartile is 87. These numbers show that students actively used the system to prepare for the exam.

The daily usage depicts an expected trend: few students and few submitted solutions at the beginning of the formative assessment period, with increasing numbers during the time, reaching the highest values the day before the exam (67% of students with 2911 submitted solutions). A few students used the system even during the night before the exam. See Fig. 5 for the complete trend.

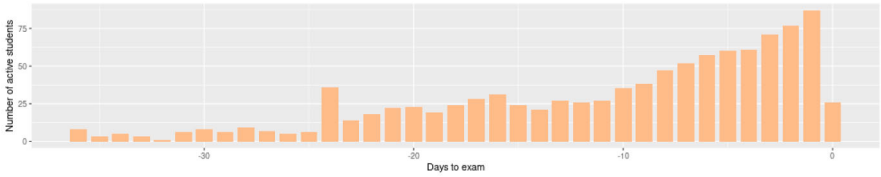
RQ$_1$

Figure 6 summarises the results of the expectation/experience analysis, for the two questions regarding the system's ability to (i) show students their learning progresses, and (ii) return helpful feedback, declined in terms of expectation (asked before interacting with the system) and experience (asked right before the exam).

The results show that 65% of students (88 over 136) answered the expectation questionnaire, and 54% (73 over 136) answered the experience questionnaire. This



(a) Number of submitted solutions by the days left for the exam.



(b) Number of active students by the days left for the exam.

Fig. 5 System usage during formative assessment period

reduction was expected because the experience questionnaire was submitted close to the exam, when the students probably preferred focusing on studying the subject rather than answering a survey. The histograms on top highlight that the distribution of answers moves towards a more positive experience (the cyan bars) than the expectation (the red bars). This is true for both questions. For instance, focusing on the highly positive cases:

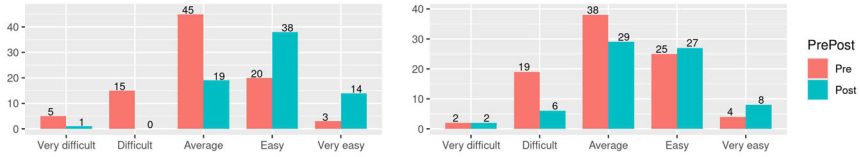
- Only 3 out of 88 students (3.4%) expected that understanding their learning progress could have been very easy. After the formative assessment phase, 14 out of 73 students (19.2%) reported that they could actually understand their learning progress very easily. See the last two bars of the histogram in Fig. 6-a);
- Only 4 out of 88 students (4.5%) expected to receive feedback that could be very easy to understand and use to solve the exercise correctly. After the formative assessment phase, 8 out of 73 students (11.0%) reported that the feedback was actually very easy to understand and use to solve the exercise. See the last two bars of the histogram in Fig. 6-b).

The table on the bottom of Fig. 6 summarises the distributions above, using averages with confidence intervals, for a faster comparison. It shows that we measured an increase in the average ratings from 3.01 to 3.89 for understanding the learning progress and from 3.11 to 3.47 for the feedback usefulness. Both differences are statistically significant.

In summary, these results show that the students experienced the system better than expected, could understand their progress, and received comprehensible feedback.

RQ_2

Table 2 summarises the results regarding the students' engagement. It reports on the rows the engagement in general (UAES), for the focused attention (UAES.FA), perceived usability (UAES.PU), aesthetic (UAES.AE), and rewarding (UAES.RW). The table on the left concerns the number of responding students and the average



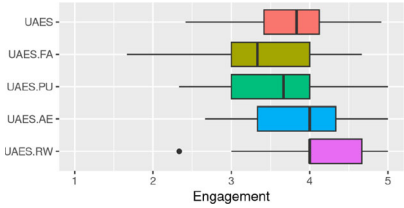
(a) Understanding progress (b) Helpful feedback

	Expectation		Experience		
	N	Mean with 95% CI	N	Mean with 95% CI	p-value
Understanding	88	3.01 [2.83, 3.20]	73	3.89 [3.71, 4.07]	< 0.01
Feedback	88	3.11 [2.93, 3.30]	73	3.46 [3.25, 3.67]	0.004

Fig. 6 The histograms on top show the frequency distribution of the answers, in red for the expectation, in cyan for the experience. The expectation/experience table summarises the number of responding students, the average answer (with the 95% confidence intervals) for the expectation and the experience, and the p-value of the paired t-test

Table 2 Results regarding RQ_2: engagement is in general satisfactory; the lowest value is for the focused attention; the highest is for the rewarding

	N	Mean with 95% CI
UAES	43	3.79 [3.61, 3.96]
UAES.FA	43	3.46 [3.24, 3.67]
UAES.PU	43	3.66 [3.43, 3.89]
UAES.AE	43	3.91 [3.71, 4.12]
UAES.RW	43	4.12 [3.92, 4.31]



engagement (with 95% confidence intervals). The boxplot on the right shows the corresponding distributions.

Out of 136 participants, only 32% (43 students) completed the UAES questionnaire, indicating moderate participation in feedback engagement measurement. The perceived overall engagement is satisfactory (3.79/5). The most positive result is the rewarding dimension: it has the highest average (4.12/5), and its distribution – apart from an outlier – shows that students found the system very useful. On the other hand, focused attention has the lowest average, with the most variable distribution.

Table 3 compares the results regarding the students' engagement for the two academic years 2022/23 and 2023/24, with a specific focus on the focused attention dimension and its related questions (UAES.FA.S1, UAES.FA.S2, and UAES.FA.S3). The analysis concerns the number of responding students, the average engagement (with 95% confidence intervals), and the p-value from the independent two-sample t-test.

Surprisingly, but expected given the previous results, we observed a statistically significant reduction in focused attention, and in its two questions regarding the perception of the time spent during the assessment (FA.S2, "The time I spent during

Table 3 Results regarding RQ_2: we observe a statistically significant reduction in focused attention, particularly for the second and third questions

	AY 2022/23		AY 2023/24		
	N	Mean with 95% CI	N	Mean with 95% CI	p-value
UAES	73	3.92 [3.77, 4.06]	43	3.79 [3.61, 3.96]	n.s.
UAES.FA	73	3.81 [3.67, 3.96]	43	3.46 [3.24, 3.67]	0.007
UAES.FA.S1 (-)	73	2.50 [2.27, 2.73]	43	2.81 [2.48, 3.15]	n.s.
UAES.FA.S2	73	3.90 [3.73, 4.08]	43	3.51 [3.24, 3.78]	0.018
UAES.FA.S3	73	4.04 [3.89, 4.20]	43	3.67 [3.38, 3.96]	0.028
UAES.PU	73	3.86 [3.67, 4.04]	43	3.66 [3.43, 3.89]	n.s.
UAES.AE	73	3.85 [3.66, 4.04]	43	3.91 [3.71, 4.12]	n.s.
UAES.RW	73	4.15 [3.95, 4.35]	43	4.12 [3.92, 4.31]	n.s.

The other differences are non-statistically significant. The (-) symbol indicates the question has a negative scoring, i.e., the lower, the better

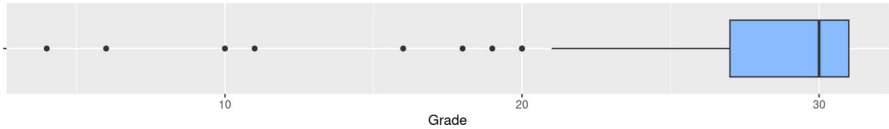


Fig. 7 Distribution of the final grades

the assessment just slipped away") and his/her involvement in the process (FA.S3, "I felt involved in the assessment"). Instead, no statistically significant difference is measured in the usability, aesthetic, and rewarding dimensions.

RQ_3

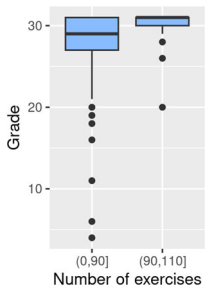
Figure 7 depicts the distribution of the final grades achieved during the first exam session by the students in the AY 2023/24 cohort. The results are generally very positive, with a few students who failed to pass the exam (i.e., the 5%).

Different engagement and the number of submitted solutions do not appear to influence grades. On the other hand, the number of attempted exercises influences the average grades: students who completed more than 90 different exercises achieved, on average, higher grades. The difference is statistically significant ($p = 0.002$). This is shown on top of Table 4: the 81 students who completed no more than 90 exercises achieved an average grading of 27.6/30; the other 25 students (who solved more than 90 exercises) obtained an average grading of 29.9/30. The distributions in the boxplot visually explain the aforementioned differences. This difference is consistently true for (almost) all numbers of attempted exercises (see Fig. 8), with differences that become statistically significant in the range (90, 97]. For higher numbers of exercises, the cardinality of students who completed more exercises reduces and, therefore, the difference between average grades reduces as well.

Table 4 The table on top reports the differences in grades with respect to the number of exercises

		N	Mean with 95% CI	p-value
Number of exercises	≤ 90	81	27.6 [26.4, 28.7]	0.002
	> 90	25	29.9 [28.9, 30.9]	

		N	Mean with 95% CI	p-value
AY	2022/23	116	27.6 [26.8, 28.5]	n.s.
	2023/24	110	28.2 [27.3, 29.1]	



The box plot on the right highlights graphically the distribution of the grades according to the threshold (i.e., > 90). The table at the bottom reports the differences in grades with respect to the two AYs

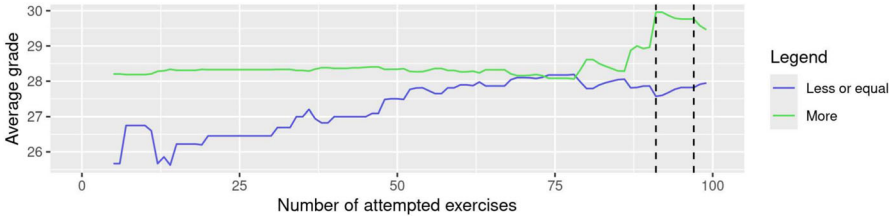


Fig. 8 Average grade by number of attempted exercises: in green/blue, the average grade of students that attempted more/no more exercises than the number indicated in the abscissa

The comparison between the two AYs 2022/23 and 2023/24 (when we introduced the feedback based on LLMs) shows an improvement in the average final grades (from 27.6/30 of the AY 2022/23 to 28.2/30 of the AY 2023/23), even if this difference is not statistically significant (see bottom of Table 4).

The final analysis regards the effect of attempting a new exercise on the students' proficiency during formative assessment. As explained in Section "Materials and Methods", we use a grade as the measure of a student's proficiency at the end of the course, and the ability as the estimated proficiency during formative assessment. Therefore, we retrieved from our system the student's ability after every exercise and calculated the corresponding increment (or decrement). On average, we observed an increase of 0.0055 (with 95% confidence intervals [0.0035, 0.0075]). This corresponds to an increase in grade of 0.028 (with 95% confidence intervals [0.018, 0.039]), calculated by applying a linear transformation of the ability (which usually ranges from −3 to 3) to the grade (which instead ranges from 0 to 31, i.e., 30 cum laude). This finding supports the previous analysis and appears intuitive: on average, attempting a new exercise enhances ability, which leads to a better grade.

Discussion

Our study shows mixed results regarding the impact of using LLMs to provide feedback to students during formative assessment in our data science courses.

In general, the students thoroughly used the system in terms of quantity (i.e., the number of submitted solutions) and quality (i.e., the number of different exercises attempted by the students), with increased intensity as the exam approached. The expectation of the students was fair regarding the ability of the system to provide clear insights into their understanding of the subject and effective feedback to support them in solving the exercises. In contrast, their experience was very good in absolute terms and better than expected, representing an excellent result for our project. Among the results, the students' opinions regarding the perceived usefulness of our LLM-based feedback (the *combined feedback*) are the most relevant to our research because they directly address the first research question. This finding aligns with related studies (Ruwe & Mayweg-Paus, 2024; Zhang et al., 2024; Lyu et al., 2024) that highlight that students perceived LLM feedback as reliable or even more trustworthy than a teacher.

A numerical comparison can be performed with the work of Gabbay and Cohen (2024). The authors used a 5-point Likert scale (similar to ours) about the utility of the feedback to explain the mistakes, measured an average rating of 2.98/5, whereas we reached an average rating of 3.46/5 (see Fig. 6). This difference is statistically significant (One Sample t-test, p-value < 0.001).

The analysis of the engagement shows unexpected results. Students reported difficulties in focused attention, which we found significantly lower than those of the AY 2022/23 students. The reason could be that the *schematic feedback* focuses on the command and how it should be modified to solve the exercise correctly, without giving additional context about the statistical background and the reason for the mistake. On the other hand, the *combined feedback* provides more text and explanations that require working memory and, therefore, additional cognitive effort to be understood and put into practice. Figure 1 shows the two feedback structures and the text returned by the two different approaches, where, e.g., the LLM even explains which test a student should potentially issue, depending on the result of the normality tests. It is worth remarking on a study Stadler et al. (2024) that found a different result: using an LLM instead of searching the web to derive valid recommendations and justifications on a given topic leads to a decreased cognitive load. These contrasting results are supported by a recent literature review (Lo et al., 2024), whose findings indicate weak (cognitive) engagement and disengagement in LLM-supported learning (e.g., over-dependence, diminished intellectual engagement, and hindering students' critical thinking abilities were the major identified weaknesses). However, this point remains unclear: a different review (Zhang & Tur, 2024) lists studies showing that an LLM can promote critical thinking and enhance problem-solving abilities — provided that diverse and challenging assignments are created.

Focusing on the last request question, the grades of students of the current cohort (i.e., those who used the *combined feedback*) improved, even if we did not measure a statistically significant difference. This means that the observed difference could be due to random chance, i.e., it is not strong enough to be likely attributed to the adoption of the *combined feedback*. On the other hand, the students' average grades in the 2022/23 cohort were already high since the *schematic feedback* could already provide effective feedback to students: previously published research (Vittorini, 2023) showed that introducing AI feedback (and gradual improvements, starting from no computer-based feedback) increased engagement and grades. Therefore, this "ceiling effect" [4] reduces the possibility of measuring a statistically significant difference (if any) without, e.g., substantially increasing the experimental cohort's numerousness. In addition, the results from the literature focusing on the learning outcomes (Meyer et al., 2024; Lyu et al., 2024) are sharper than ours, probably because the authors compared the effect of LLM feedback with no (computer-based) feedback. Comparing the average grades achieved by students who used the *combined feedback* with students who used no automated feedback leads to a statistically significant difference: the

[4] A ceiling effect is defined as the property of the distribution of data that accumulates toward the high end.

average grade in 2020 was equal to 23.5, and a comparison with the grades of AY 2023/24 yields a statistically significant difference with a p-value < 0.001.

Finally, it is worth mentioning that — in our study — we did not experience hallucinations during the system development and testing, nor did students during formative assessment activities (or at least they were unable to recognize them), like those reported in Gabbay and Cohen (2024). However, students who participated in the study voluntarily reported possible improvements. The most frequent issue they asked to address was related to the suggestions given by the LLMs, which sometimes included commands not discussed during the lectures. For example, to sum the values of some dataset's columns, we used the simple "+" arithmetic operator in the course, whereas the LLM suggested using the `rowSums` function.

Conclusions, Limitations and Future Work

This study summarized an attempt to use LLMs to enhance feedback in data science education. The research reveals that the *combined feedback*, i.e., LLM-generated feedback combined with traditional methods, if compared with a traditional AI feedback, did not yield clear benefits in terms of learning outcomes. Conversely, the results coming from the perceived usefulness of the feedback, the engagement, and the overall grades (when compared with a cohort that did not use any automated feedback) suggest a possible positive effect of adopting an LLM-based solution to support student feedback, even if this solution seems to require additional cognitive effort to students. This latter point needs further research, particularly in light of the contrasting results reported in the literature (Lo et al., 2024; Zhang & Tur, 2024). In addition, we observed that students who solved many different exercises achieved, on average, better grades than the others, with statistically significant differences when the number of exercises exceeded 90. This result – which could be in general expected (i.e., students who exercise more perform better) – suggests pushing students to try exercises, in our case, until approaching this threshold: this strategy can be implemented both during lectures and/or through the system, e.g., with reminders or push notifications.

In summary, the results suggest that LLM-enhanced feedback, if compared with traditional AI feedback, does not seem to increase the learning outcomes, and requires additional cognitive effort for students. On the other hand, students appreciated the LLM-enhanced feedback (more than expected) and thoroughly used the system for their preparation, with students who solved more different exercises achieving better grades.

This study has the following limitations. First, we compared cohorts from different academic years. Even if the two cohorts have comparable numerousness, similar gender distribution, belong to the same courses, and passed an admission test that establishes a minimum level of competence, the absence of randomization in the cohorts may still represent an issue. It is normal to observe fluctuations in grades between different years, even if there are no changes in the composition of the students' cohorts. Second, the measure regarding the cognitive load is a subscale of the engagement and, therefore, does not analyze the reasons behind this finding deeply.

Further future research directions include the following:

- using more detailed measures (e.g., Ouwehand et al., 2021) to improve the understanding of how the *combined feedback* affects students' cognitive load;
- refining the prompt used to generate the feedback and adopting Retrieval-Augmented Generation (RAG) techniques (Lewis et al., 2020), with the twofold aim of:
  - improving focused attention (e.g., by reducing verbosity);
  - better aligning the feedback with the course content (i.e., to tackle the students' suggestions).
- evaluating the possibility of introducing a pedagogical agent like in Lang et al. (2022), to maximize learning, acting on students' emotions and intrinsic motivation, to reduce the cognitive load;
- conducting longitudinal studies to assess the long-term impact on the learning outcomes.

In conclusion, this research adds further pieces of evidence about the potential and the limits of LLMs to provide effective feedback to students. It raises interesting questions about the cognitive effort required by students when dealing with LLM-generated feedback and opens future research directions.

A Appendix

A.1 Expectation/experience questionnaire

In a scale from 1 to 5, where 1=Very difficult, 2=Difficult, 3=Average, 4=Easy, and 5=Very easy, please answer the following questions:		1	2	3	4	5
Expectation						
PRE-1	Using the system, how easy do you expect it will be to understand your progress and level of preparation in the subject?					
PRE-2	Using the system, how easy do you expect it will be to understand and correct your mistakes to arrive at the solution of an exercise?					
Experience						
POST-1	Using the system, how easy had it been to understand your progress and level of preparation in the subject?					
POST-2	Using the system, how easy had it been to understand and correct your mistakes to arrive at the solution of an exercise?					

A.2 UAES

In a scale from 1 to 5, where 1=completely disagree, 2=disagree, 3=neutral, 4=agree, and 5=completely agree, please answer the following questions:		1	2	3	4	5
UAES.FA : Focused attention						
FA-S1	I felt lost during the assessment (neg)					
FA-S2	The time I spent during the assessment just slipped away					
FA-S3	I felt involved in the assessment					
UAES.PU : Perceived usability						
PU-S1	I felt frustrated during the assessment (neg)					
PU-S2	I found the assessment confusing (neg)					
PU-S3	The assessment carried out in this way was taxing					
UAES.AE : Aesthetic appeal						
AE-S1	The system was attractive					
AE-S2	The system was aesthetically appealing					
AE-S3	The system was captivating					
UAES.RW : Rewarding						
RW-S1	Using the system was worthwhile					
RW-S2	Performing the assessment in this way was rewarding					
RW-S3	I felt interested in the assessment					

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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